0.31 0.31 Ans

$\Rightarrow 97.4\%$

(3) $P(G) = 0.1 \rightarrow$ Gold present

$P(C) = 0.3 \rightarrow$ coal present

$P(N) = 0.6 \rightarrow$ Neither

$P(+|G) = 0.8$

$P(+|C) = 0.4$

$P(+|N) = 0.2$

$P(G|+) = ?$

$P(+) = P(+|G)\,P(G) + P(+|C)\,P(C) + P(+|N)\,P(N)$

$\Rightarrow (0.8 \times 0.1) + (0.4 \times 0.3) + (0.2 \times 0.6)$

$P(+) \Rightarrow 0.32$

$P(G|+) = \dfrac{0.8 \times 0.1}{0.32} \Rightarrow \dfrac{0.08}{0.32} = 0.25$

probability of gold is present in positive test

$\Rightarrow 0.25\ (25\%)$ Ans